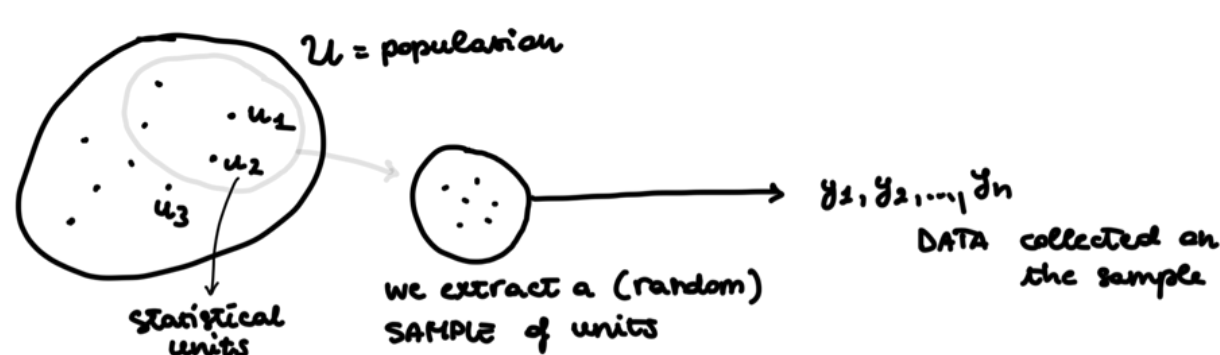


PREREQUISITES OF STATISTICAL INFERENCE



The observations are $y_i = y(u_i)$

STATISTICAL INFERENCE: we try to understand the population starting from a sample

We use probabilistic tools

STATISTICAL MODEL: on the data, we specify a probabilistic model suited to describe a particular phenomenon.

probabilistic model: $Y \sim p_\theta(y)$ we draw $(y_1, \dots, y_n)$

statistical model: given $(y_1, \dots, y_n)$, we define a set of "reasonable" distributions $p(y)$ that could have generated it, and try to recover the particular $p_\theta(y)$ within this set

$$y_i \rightsquigarrow Y \sim p(y; \theta) \quad \theta \in \Theta \subseteq \mathbb{R}^p$$

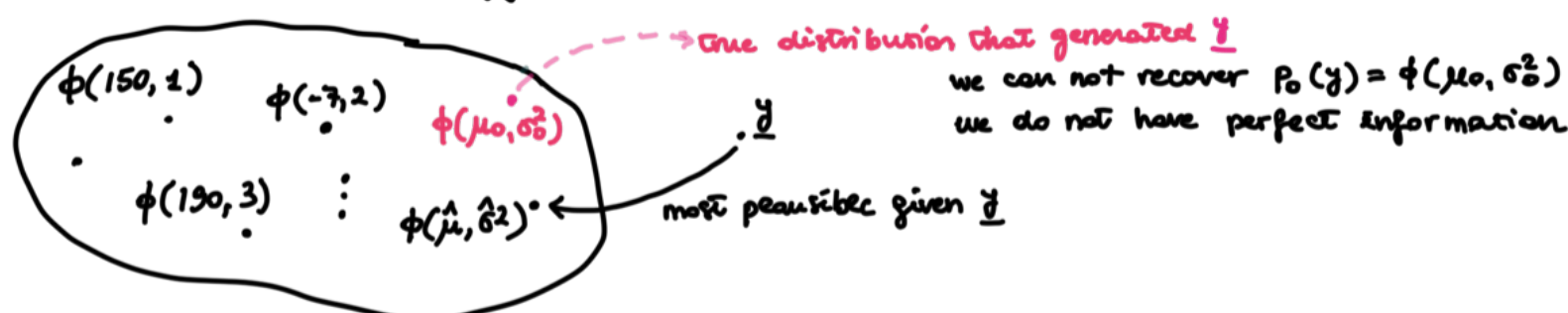
known (prob. model) unknown (parameter); identifies the particular element

e.g. $y = (y_1, \dots, y_n)$ heights of a sample of students
a Gaussian distribution is a reasonable model

$\Rightarrow$ statistical model $Y_i \sim N(\mu, \sigma^2)$ random variable: before observing the data. The distribution of Y describes all possible realizations y_i
 $\mu \in \mathbb{R} \quad \sigma^2 \in (0, +\infty)$

MODEL: set of all Gaussian distributions

inference: we use the data to identify the element in this set that best describes them.



e.g. $y = (y_1, \dots, y_n)$ counts of cars passing in a street
statistical model $Y_i \sim \text{Pois}(\lambda) \quad \lambda \in (0, +\infty)$ indep.

Methodologies:

- POINT ESTIMATION: we identify one element (the most plausible) within the set of distributions $\hat{p}(y)$
- INTERVAL ESTIMATION (confidence intervals): subset of reasonable elements, where the subset has a known degree of "confidence" that the true element (p_θ) is contained in it.
- HYPOTHESIS TESTING: we ask if there is enough evidence in the sample to draw conclusions about a particular statement ("null hypothesis")

POINT ESTIMATE

given the set of elements $\{p(y; \theta) ; \theta \in \Theta \subseteq \mathbb{R}^p\}$ we want to identify the most plausible $\hat{p}(y)$

It is identified through a particular element of Θ : $\hat{\theta}$ i.e. $\hat{p}(y) = p(y; \hat{\theta})$

ESTIMATE $\hat{\theta} = \hat{\theta}(y)$ is a function of the observed values (realizations)

ESTIMATOR $\hat{\Theta} = \hat{\Theta}(Y)$ is a function of the random variable

$\Rightarrow$ we study the distribution of the estimator, its expected value, variance, ...

HYPOTHESIS TESTING

$$\begin{cases} H_0: \theta \in \Theta_0 \subset \Theta & \text{null hyp.} \\ H_1: \theta \in \Theta \setminus \Theta_0 & \text{alternative hyp.} \end{cases} \quad \text{e.g.} \quad \begin{cases} H_0: \theta = \theta_0 \\ H_1: \theta \neq \theta_0 \end{cases}$$

TEST: we partition the sample space into the REJECT (R) and ACCEPTANCE (A) regions

A : values of y that suggest that H_0 is true

R : values of y that suggest that H_0 is false $\rightarrow$ reject H_0

TEST STATISTIC: a function of the data that defines the two regions. $T(y) = t^{obs}$

$A = \{y \in \mathcal{Y} : T(y) \text{ suggests } H_0\}$

$R = \{y \in \mathcal{Y} : T(y) \text{ suggests } H_1\}$

Before looking at the data y , T is a function of the random variable $Y \rightarrow T(Y)$ is a random variable

How do we draw conclusions?

I. FIXED SIGNIFICANCE LEVEL α

We guard against the 1st type error: we fix α to a small value (e.g. $\alpha = 0.01$, $\alpha = 0.05$, $\alpha = 0.10$)

and we derive A and R so that $IP(\text{reject } H_0 \mid H_0 \text{ is true})$ is equal to α (or at most α)

In other words, $\alpha = IP(1^{\text{st}} \text{ type error}) = IP(Y \in R \mid H_0 \text{ true})$

Of course, the smaller α is, the smaller R will be (I want to reject H_0 only if I am really really confident)

II. OBSERVED SIGNIFICANCE LEVEL (p-value) α^{obs}

it is the probability of observing "more extreme" values (i.e. more against H_0) than the ones we observed.

• if the reject region is of a one-tailed test

$$H_1: \theta > \theta_0 \quad (\text{right tail}) \Rightarrow \alpha^{obs} = IP_{\theta_0}(T \geq t^{obs}) = IP(T(Y) \geq t(y^{obs}) \mid H_0 \text{ true})$$

$$H_1: \theta < \theta_0 \quad (\text{left tail}) \Rightarrow \alpha^{obs} = IP_{\theta_0}(T \leq t^{obs})$$

• if the reject region is of a two-tailed test

$$H_1: \theta \neq \theta_0 \Rightarrow \alpha^{obs} = 2 \min\{IP_{\theta_0}(T \geq t^{obs}); IP_{\theta_0}(T \leq t^{obs})\}$$

The two procedures are related: if $\alpha^{obs} < \alpha$, then I reject H_0 in a fixed-level test of level α

CONFIDENCE INTERVALS of confidence $(1-\alpha)$

it is a random interval $\hat{C}(Y)$ such that $IP(\theta \in \hat{C}(Y)) = 1-\alpha$ for all $\theta \in \Theta$

With probability $(1-\alpha)$, the interval contains the true value of the parameter, whatever it is.

After we compute the interval with the data (hence, we get a fixed numeric interval), it either contains the true θ or not.

The probability must be interpreted regarding to the random quantity.

We build it through the identification of a PIVOTAL QUANTITY: a function $g(Y; \theta)$ of the r.v. Y and the parameter θ such that its distribution does not depend on θ (hence it is completely known).

Then we look for the interval (u, v) such that $1-\alpha = IP(u < g(Y; \theta) < v)$.

With the data we compute $\hat{C}(y^{obs}) = \{\theta \in \Theta : g(y^{obs}, \theta) \in (u, v)\}$

$\hookrightarrow$ all values of the parameter that, given the observed data, give a value of g within the (u, v) interval.